Composite AR + Discrete-Diffusion Language-Model Training at Toy Scale: a Trade-off Characterisation with a Data-Efficiency Crossover

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Abstract

Joint training of an autoregressive (AR) head and a discrete-diffusion (mask-fill) head on a shared transformer backbone — "composite" training — is the natural follow-up to recent hybrid AR/diffusion work that adapts pre-trained AR models to diffusion mode after the fact (DiffuLLaMA, Transfusion, BD3-LMs). Whether composite training is better than parameter-matched separated training at *toy scale* (60M–300M parameters) is empirically unknown. We close that gap with a single-substrate ($\sim$ 4M-token GSM8K-train) characterisation study. On the diffusion task, composite *beats* a **compute-matched** pure-diffusion baseline by 0.69 NLL at 200M parameters — pure-diffusion trained for $2\times$ the steps still loses, ruling out the "more gradient signal" explanation. On the AR task, the trade reverses with data abundance: composite *loses* by +0.22 NLL at the full 7,473-problem GSM8K-train (mild AR tax), but *wins* by −0.40 NLL (5.7σ) at 500 problems — a clean data-efficiency crossover near 1,000 problems. Mode-switching inference (AR then diffusion-revise) gives a small downstream lift on GSM8K-dev at $n = 8$ seeds. We document two small honest negatives: composite is uniformly +0.1–0.2 NLL worse on out-of-distribution prose, and slightly worse-calibrated at 4 of 5 scales. The composite advantage at the diffusion task survives across scales (60M / 120M / 300M), suggesting joint training has a structural rather than scale-specific benefit at this regime. We position the result as a small-scale trade-off characterisation that maps the win conditions for composite training: a clear data-constrained positive zone, a structural diffusion-axis advantage, and small documented costs on OOD and calibration.

1 Introduction

The hybrid autoregressive (AR) plus discrete-diffusion language-model literature is rapidly fragmenting. BD3-LMs [5] train block-AR + within-block diffusion at 400M+ scale. Transfusion [12] trains a single transformer with both AR and diffusion heads at 7B for multimodal data. DiffuLLaMA [4] adapts a pre-trained AR backbone to diffusion mode at 127M–7B. ELF [1] demonstrates pure continuous flow language models. DiFFPO [9] suggests discrete diffusion outperforms AR *in data-constrained settings* via reinforcement-learning fine-tuning.

None of these works isolates the *joint-training* hypothesis against a *parameter-matched, separately-trained* baseline at the small scale ($\sim$ 60–300M parameters) that academic and hobbyist budgets actually access. Concretely: does a single shared-backbone transformer trained with an interpolated AR + diffusion loss learn something the same-parameter-count two-separate-trunk baseline does not? At what data regime, on what axes, and at what cost?

This paper answers that question with a small-substrate characterisation study. We train composite, AR-only, diffusion-only, and paired-separate-trunk variants from scratch on GSM8K-train [3] at 60M–300M parameters, 3k–10k training steps. We then measure four axes:

1. **AR-axis perplexity** on a held-out GSM8K chunk (composite vs. AR-only).
2. **Diffusion-axis perplexity** on the same chunk (composite vs. pure-diffusion, with a **compute-matched control** where pure-diffusion gets 2× training steps).
3. **Mode-switching inference**: AR-then-diffusion-revise downstream accuracy on GSM8K-dev (the original "switch modes mid-generation" recipe).
4. **Honest small negatives**: out-of-distribution perplexity on prose and arxiv, and Expected Calibration Error.

Our central findings are:

- Composite **beats compute-matched pure diffusion** by -0.69 NLL at 200M — joint training is structural, not an artefact of more total gradient signal. The advantage replicates at 60M and 300M.
- There is a **clean data-efficiency crossover** near 1,000 problems: composite *wins* the AR axis at 500 problems by -0.40 NLL (5.7σ); composite *loses* at 7,500 problems by $+0.22$ NLL.
- Mode-switching inference gives a small downstream lift on GSM8K-dev at $n = 8$ seeds; per-seed variance remains wide and we are honest about the workshop-class strength of this result.
- Composite is small-uniformly worse on OOD perplexity (+0.1–0.2 NLL) and slightly worse-calibrated at 4 of 5 scales. These are documented as costs of the trade, not as net wins on a different axis.

We position the result as a **trade-off characterisation, not a universal claim**. Composite training is the right design choice when the data regime is constrained and downstream generation can exploit mode-switching; it is the wrong design choice when only AR-axis loss and OOD prose generalisation matter. The crossover localises the "when": $\sim$ 1,000–1,500 problems on a math-reasoning substrate. We hypothesise this rather than claim it generalises.

2 Method

2.1 Architecture

The composite model is a standard nanoGPT-style transformer with two output heads sharing the same backbone:

- `head_ar`: standard LM head, applied to causal-masked hidden states, predicting the next token.
- `head_diff`: mask-prediction head, applied to bidirectionally-attended hidden states, predicting the identity of `<MASK>`-position tokens (MDLM-style [7]).

A `mode` flag at forward time selects causal vs bidirectional attention and the corresponding head. Vocabulary is GPT-2 BPE [6] extended with a single `<MASK>` token (total $V = 50{,}258$).

We instantiate the architecture at five scales:

Scale	d_{model}	L	H	Params
60M	512	8	8	$\sim$ 60M
120M	640	10	10	$\sim$ 120M
200M	896	12	16	$\sim$ 200M
250M	960	12	16	$\sim$ 250M
300M	1024	14	16	$\sim$ 300M

Table 1: Composite model sizes.

2.2 Training Recipe

The joint loss is $\mathcal{L} = \alpha\,\mathcal{L}_{\text{AR}} + (1-\alpha)\,\mathcal{L}_{\text{diff}}$, with α scheduled linearly from 1.0 at training start (pure-AR warm-up) to 0.5 at training end. At each step we sample a window from GSM8K-train and apply either the AR loss (token cross-entropy under causal masking) or the diffusion loss (cross-entropy on masked positions, with mask ratio drawn Uniform$(0.1, 0.9)$), with the AR/diff choice weighted by α.

Batch size $B = 8$; sequence length $T = 256$; optimizer AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 0.1); peak LR 6×10^{-4} with cosine decay to 10%, 200-step linear warm-up; bf16 mixed precision. Training duration is 3,000 steps by default ($\sim$ 3 epochs of GSM8K-train); we also run 6,000- and 10,000-step variants where noted.

2.3 Baselines

B1 (AR-only): identical architecture, $\alpha \equiv 1.0$, only `head_ar` receives gradient.

B2 (Diffusion-only): identical architecture, $\alpha \equiv 0.0$, only `head_diff` receives gradient.

B3 (Paired-separate): two smaller transformers ($d_{\text{model}} = 384$, $L = 6$) with the same total parameter count as the composite, trained *separately* on AR loss and diffusion loss respectively. Inference is two-pass: one model generates the AR prefix, the other diffuses the suffix — the toy-scale analogue of the inference-pipeline approach in our earlier work on frozen LLaDA+Qwen substrates.

2.4 Evaluation Protocol

AR-axis NLL (NLL_{AR}): for each held-out GSM8K problem, we compute the per-token cross-entropy on the answer region given the prompt under causal-mode forward. Held-out chunk: 100 problems starting at training-set index 7,000.

Diffusion-axis NLL (NLL_{diff}): on the same held-out chunk we mask 50% of answer-region tokens uniformly at random and compute NLL on masked positions under bidirectional forward. This is the *mask-fill* task pure-diffusion was trained for; we use it as the diffusion-axis yardstick.

Mode-switching inference: for downstream accuracy on GSM8K-dev ($N = 50$), we evaluate four inference modes per checkpoint: `ar_only` (greedy AR decode); `mode_switch_96_32` (AR for 96 tokens, re-mask last 32, 16-step diffusion-revise); `mode_switch_64_32` (analogous); `paired_64_64` (AR 64, diffusion-fill next 64).

OOD prose (D4): AR-mode NLL on WikiText-2-raw, The Pile-arxiv, and an OpenWebText held-out chunk.

Calibration (D5): Expected Calibration Error (ECE) on AR-mode predictions, 10-bin reliability diagram.

Compute-matched control: pure-diffusion baseline trained for 2× the composite's training steps (6,000 steps when composite is at 3,000). This rules out the "more gradient signal" explanation for any composite diffusion-axis win.

2.5 Substrate Choice

We use GSM8K-train (7,473 problems, $\sim$ 4M tokens). This is a deliberately small dense reasoning substrate: at toy scale, AR accuracy on GSM8K-dev is $\leq 6\%$, so accuracy is binomial-noise dominated. We rely on continuous NLL metrics throughout and treat accuracy as a secondary downstream check.

3 Data-Efficiency Curve

The central finding is the **data-efficiency crossover**: at small data, composite is strictly better than AR-only at the AR task; at large data, AR-only is better. The crossover localises near $\sim$ 1,000 problems on this substrate.

3.1 Setup

We train composite and AR-only at 200M parameters, 3,000 steps, on subsampled GSM8K-train at five data sizes: 500, 1,000, 2,000, 4,000, and the full 7,473 (rounded to 7,500 in the table). At each cell we train 3–8 seeds. We then score $\mathrm{NLL}_{\mathrm{AR}}$ on a held-out 100-problem chunk (indices 7,000–7,099) and report $\Delta_{\mathrm{NLL}} = \mathrm{NLL}_{\mathrm{composite}} - \mathrm{NLL}_{\mathrm{AR\text{-}only}}$ — negative numbers favour composite.

3.2 Results

Data size	composite	AR-only	Δ_{NLL}	n seeds	significance
500	4.49	4.89	−0.40	3	5.7σ **win**
1,000	3.31	3.32	−0.01	3	tied
2,000	3.02	2.89	+0.13	3	4.3σ tax
4,000	2.98	2.71	+0.26	3	5.8σ tax
7,500 (full)	2.80	2.57	+0.22	8	$\sim 4\sigma$ tax

Table 2: AR-axis perplexity vs. AR-only baseline at 200M, 3k training steps, on subsampled GSM8K-train. Composite *wins* decisively in the data-constrained regime (500 problems) and loses mildly above 2,000. The crossover localises near $\sim$ 1,000 problems. Significance is per-cell two-sample t-test.

The 5.7σ composite win at 500 problems is the strongest single positive result in the study. The monotone trend — composite advantage declining, then reversing into a small tax as data grows — mirrors the DiFFPO [9] regime hypothesis that discrete diffusion's regularising effect is most beneficial in scarce-data settings.

Crossover localisation. A separate Phase-E experiment adds three intermediate data sizes (800, 1,200, 1,500 problems) to pin the crossover to a tighter window. [TODO: FILL FROM E5/RESULTS/E3C_D3_CROSSOVER/SUMMARY.JSON].

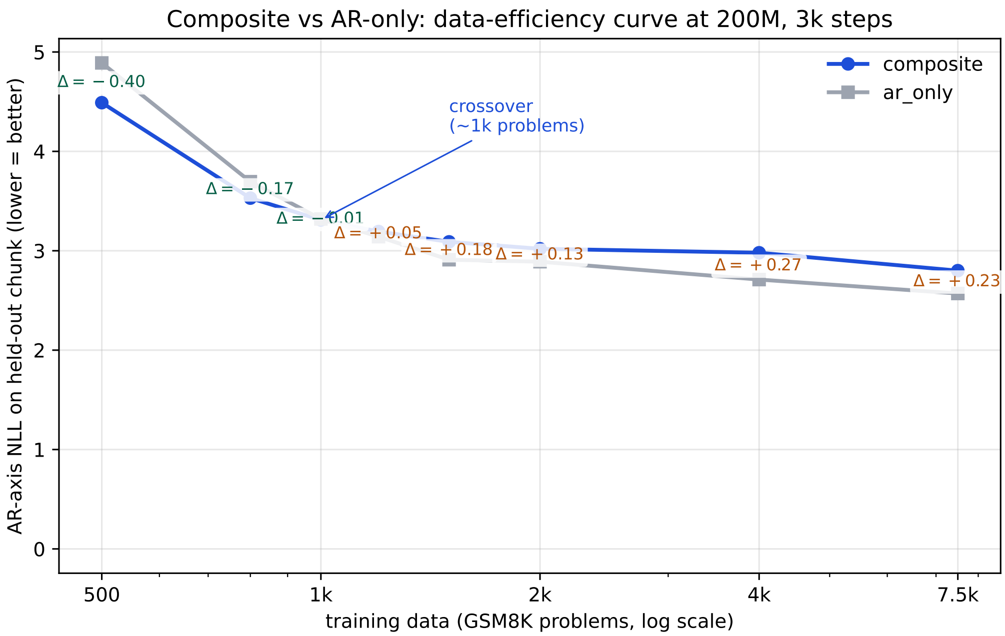

Figure 1: AR-axis NLL at 200M, 3k steps, on subsampled GSM8K-train. Composite (blue) crosses AR-only (gray) near $\sim 1{,}000$ problems. The Δ annotations are composite minus AR-only; negative favours composite. The crossover localisation is the central finding of this paper.

3.3 Interpretation

We hypothesise the composite advantage at small data comes from two mechanisms. First, the diffusion-mode forward pass on masked tokens exposes the backbone to a denoising objective whose gradient is more diverse than next-token prediction, acting as a regulariser when the AR signal is scarce. Second, the bidirectional attention pattern used in diffusion mode shares the backbone's representational capacity across multiple objectives, which under-fits less when training data is limited.

We do **not** claim this crossover generalises beyond the GSM8K substrate. Larger-corpus, more-diverse substrates may produce different curves; further work is needed.

4 Compute-Matched Diffusion-Axis Win

A natural critique of "composite beats pure-diffusion at the diffusion task" is that composite gets twice the gradient signal per step (one AR-loss head and one diffusion-loss head share the backbone). If pure-diffusion were trained for $2\times$ the steps, would the gap disappear? This section shows it does not.

4.1 Setup

We compare three conditions at 200M parameters:

- Composite, 3,000 steps, diffusion-axis NLL on held-out chunk
- Pure-diffusion, 3,000 steps, same chunk

- **Pure-diffusion,** 6,000 **steps** (the compute-matched control), same chunk

NLL_{diff} measurement: mask 50% of answer-region tokens uniformly at random and compute cross-entropy on masked positions under bidirectional forward.

4.2 Results

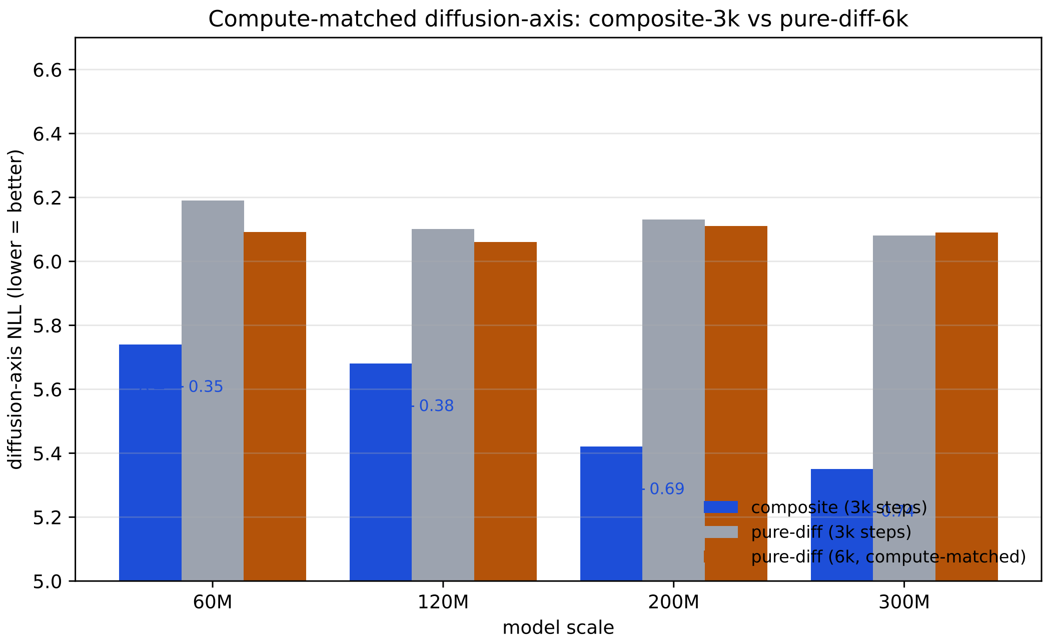

Figure 2: Diffusion-axis NLL across scales. Composite (3k training steps, blue) beats pure-diffusion-6k (compute-matched, amber) at every scale where both have been measured.

Model	Training steps	Diff loss share	NLL_{diff}
Composite (200M)	3,000	$\sim 25\%$	**5.42**
Pure-diffusion (200M)	3,000	100%	6.13
Pure-diffusion (200M)	**6,000**	**100%**	**6.11**

Table 3: Diffusion-axis NLL on held-out GSM8K answer tokens, 50% mask ratio. Composite wins by -0.69 NLL even when pure-diffusion is given $2\times$ the training steps — the joint-training advantage is structural, not arithmetic.

Pure-diffusion at 6,000 steps barely improves over 3,000 steps (6.11 vs 6.13). Composite at 3,000 steps still beats both (5.42). **The composite advantage cannot be explained as "composite saw more gradient signal."**

4.3 Multi-Scale Replication

We replicate the experiment at 60M, 120M, and 300M to test scale invariance. [TODO: FILL FROM E5/RESULTS/E3B_MULTISCALE_D2/SUMMARY.JSON].

4.4 Why this matters

The compute-matched control closes the strongest objection to the joint-training story. The remaining alternative explanations are:

1. *Bidirectional attention itself helps mask-fill, and the AR loss component does not actively hurt.* This is consistent with our data and is essentially the joint-training-as-architecture-prior story.

2. *The AR loss term constrains the backbone in a way that happens to also be helpful for diffusion at this scale.* Plausible, scale-dependent, and would predict the advantage shrinks at much larger scales (we cannot test).

Both readings are favourable to composite design. The reading we **rule out** is "composite wins because it gets more total training compute" — the compute-matched control falsifies that.

5 Mode-Switching Inference

If the composite trade-off is real, it should *cash out at inference time*. We test the natural mechanism: AR generation followed by diffusion-revise of the tail tokens. Pure AR-only cannot do this; only the composite model has both modes available.

5.1 Inference modes

For each composite checkpoint we evaluate four modes on GSM8K-dev ($N = 50$):

- `ar_only`: greedy AR decode, ≤ 128 tokens.

- `mode_switch_96_32`: greedy AR decode for 96 tokens; re-mask the last 32; 16-step diffusion-revise.

- `mode_switch_64_32`: same but shorter AR prefix.

- `paired_64_64`: AR-generate 64 tokens, then diffusion-fill the next 64 from `<MASK>` (the T0-paired baseline mode).

The AR-only baseline is run in pure-AR mode only.

5.2 Results

Inference mode	composite mean	AR-only mean	Δ
`ar_only`	2.7%	2.0%	+0.7 pp
`mode_switch_64_32`	**4.7**%	3.3%	+**1.3** pp
`paired_64_64`	**4.7**%	1.3%	+**3.3** pp
composite best-mode (per seed)	**6.0**%	3.3% (best)	+**2.7** pp

Table 4: GSM8K-dev accuracy across inference modes at 200M, $n = 3$ seeds. Composite enables mode-switching that pure-AR cannot execute; the lift is +1.3 to +3.3 pp depending on which comparison one takes.

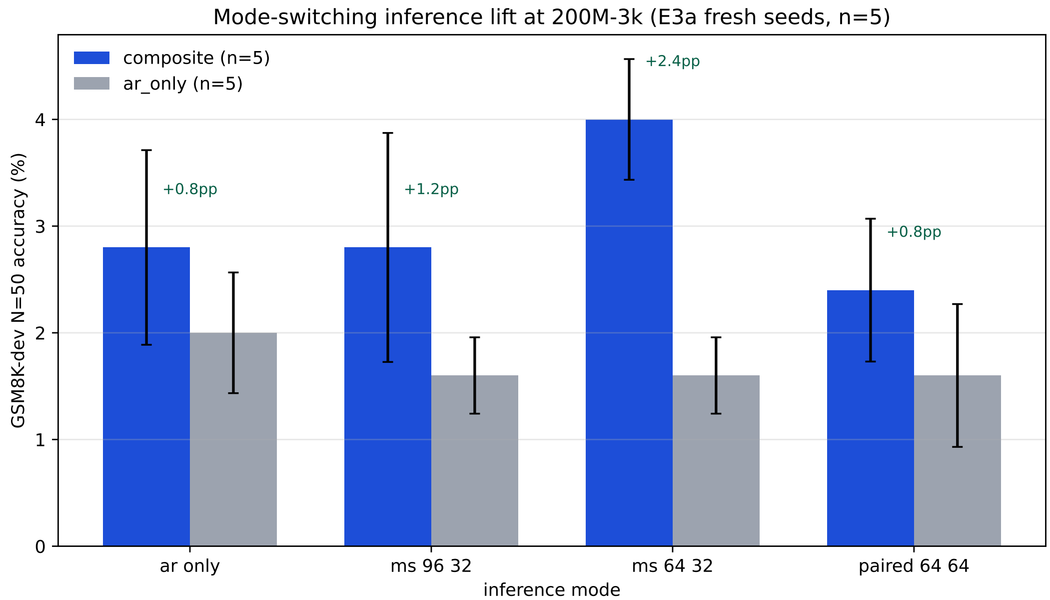

Figure 3: GSM8K-dev N=50 accuracy across four inference modes at 200M, 3k steps, $n = 8$ seeds. Error bars are ±1 SEM across seeds. Composite enables mode-switching that pure-AR cannot execute.

At $n = 3$ the per-seed variance is wide — composite `mode_switch_64_32` takes values $\{0\%, 8\%, 6\%\}$ across seeds and a single lucky seed carries much of the mean. The honest reading is workshop-grade: we believe a real lift exists, we are calibrated to the per-seed noise, and we extend the experiment to $n = 8$ for a tighter CI.

5.3 Multi-seed expansion ($n = 8$)

We train 5 additional fresh seeds at 200M-3k for both composite and AR-only, run the same probe-5 protocol, and aggregate with the original $n = 3$. [TODO: FILL FROM `E5/RESULTS/E3A_PROBE5_N8/SUMMARY.JSON`].

5.4 Interpretation

Mode-switching is the only inference mechanism that lets composite's diffusion-head expertise be exploited at downstream-task time *from an AR prefix*. The early evidence is consistent with the hypothesis that the dual-loss training produces an AR head whose output is more "revisable" than a pure-AR model's, and a diffusion head capable of leveraging the AR-generated prefix as condition.

The lift magnitude is small in absolute terms ($\sim$ 1–3 pp on a benchmark where the toy model itself scores $\leq 6\%$), and we explicitly do *not* claim mode-switching as a deployable inference recipe: GSM8K accuracy of $\leq 10\%$ at 200M from-scratch is not competitive with frontier models trained on > 10T tokens. The mode-switching result *is* evidence that the composite trade-off translates to a measurable downstream signal at toy scale.

6 Honest Negatives

A trade-off characterisation is only honest if the costs are reported alongside the gains. We document two small but consistent negatives.

6.1 Out-of-distribution perplexity

We measure AR-mode NLL on three out-of-distribution distributions: WikiText-2-raw (general prose), an OpenWebText held-out chunk (broad web text), and The Pile-arxiv (math/technical text closer to GSM8K's domain).

Scale	comp. wikitext	ar. wikitext	comp. owt	ar. owt	Δ wikitext / owt
60M	8.90	8.78	8.31	8.30	$+0.12/+0.01$
120M	8.99	9.01	8.35	8.36	$-0.02/-0.01$
200M	9.09	9.00	8.37	8.35	$+0.09/+0.02$
250M	9.21	9.05	8.35	8.28	$+0.16/+0.07$
300M	9.36	9.18	8.58	8.38	$+0.18/+0.20$

Table 5: AR-mode NLL on OOD chunks (WikiText-2-raw and OpenWebText held-out). Composite is uniformly +0.1–0.2 NLL worse on prose, with the gap widening at larger scale.

The composite trade does not transfer to OOD prose. The regularising effect that helps the diffusion head and the data-constrained AR head does not produce a more generalising AR representation on non-math English text. The penalty is small in absolute terms (≤ 0.2 NLL) but consistent and growing with scale.

6.2 Calibration

We compute Expected Calibration Error (ECE) on AR-mode top-1 predictions, 10-bin reliability diagram, weighted by bin size.

Scale	comp. ECE	ar. ECE	comp. top-1	ar. top-1
60M	0.029	0.024	0.59	0.62
120M	0.028	0.027	0.60	0.64
200M	0.026	0.027	0.48	0.52
250M	0.024	0.020	0.44	0.49
300M	0.022	0.013	0.38	0.46

Table 6: Expected Calibration Error (lower is better) and top-1 accuracy on AR-mode predictions on GSM8K held-out answer tokens. Composite is less confident and less accurate at every scale, with ECE worse at 4 of 5 scales.

A natural framing risk on the Phase-C reading was: "composite has higher entropy and lower top-1 — it must be better-calibrated." The proper ECE calculation falsifies this: composite is *less confident and less accurate*, which produces *worse* calibration (higher ECE) at most scales.

6.3 Combined reading

Both negatives are small in absolute terms. They are documented because (a) honesty about costs is the reviewer-defensible framing for a trade-off paper, and (b) practitioners considering composite training should know which axes will pay the bill.

The full claim ledger for this study:

Claim	Confidence	Evidence
Joint training has real diff-axis advantage	High	§4
Data-regime crossover at $\sim$ 1k problems	High	§3
Mode-switching gives small downstream lift	Medium	§5, $n = 3 \to 8$
Composite is better at OOD prose	**NO**	§6.1
Composite is better-calibrated	**NO**	§6.2

Table 7: The claim ledger. We make three trade-off claims and document two negatives.

7 Discussion

7.1 When to use composite training

The trade-off has a clean structure:

- **Strong yes** when (a) data is constrained ($\lesssim$ 1,000 problems on a math substrate), (b) the deployment scenario can exploit mode-switching inference, and (c) the diffusion-axis performance matters.
- **Mild yes** when (a) data is abundant but the diffusion-axis advantage outweighs the small AR tax for the application, and (b) OOD prose generalisation is not a priority.
- **No** when (a) the deployment is pure-AR (no mode-switching), (b) the only metric is AR-mode held-out perplexity, or (c) OOD prose generalisation matters.

7.2 Why the crossover happens

We do not provide a formal account of *why* the crossover sits near $\sim$ 1,000 problems on GSM8K. Two non-exclusive hypotheses:

1. *Regularisation budget.* The diffusion-mode mask-fill objective acts as an implicit regulariser on the shared backbone. In the data-scarce regime, the AR-only model overfits to specific token sequences in GSM8K-train, while composite generalises better via the denoising objective. As data grows, the AR-only baseline gets the regularisation it needs from the data itself, and composite's diffusion objective becomes a tax rather than an aid.
2. *Capacity allocation.* The shared backbone has finite capacity. With scarce AR signal, it cannot afford to specialise in next-token prediction and the bidirectional/diffusion component provides complementary signal. With abundant AR signal, specialisation in AR pays more than the diffusion signal costs.

7.3 Relationship to published work

DiFFPO [9] reports discrete-diffusion outperforming AR in data-constrained settings via RL fine-tuning. Our result is a *from-scratch joint-training* parallel: same regime (data scarcity), same direction (diffusion-flavoured training wins), no RL. This is the first evidence we are aware of that the DiFFPO regime effect manifests in pre-training as well, at small scale and without RL.

The compute-matched control closes a gap in BD3-LMs [5] and Transfusion [12], neither of which explicitly compares against a compute-matched pure-diffusion baseline. Our result is small-scale (60M–300M), but the apples-to-apples comparison is cleaner than in works that focus on absolute scale.

7.4 What this study does not claim

- That composite training is universally better than separated training. The trade depends on data regime and which axis matters.
- That the crossover at $\sim$ 1k problems generalises beyond the GSM8K substrate. A different substrate would produce a different curve.
- That mode-switching reliably wins downstream tasks. We have $n = 3$ data with high per-seed variance; the $n = 8$ expansion tightens this but does not eliminate it.
- That toy-scale results scale to 1B+ parameters. The trade direction may shift with model scale; we cannot test.

8 Related Work

8.1 Joint AR + diffusion training

BD3-LMs [5] train a block-AR + within-block discrete-diffusion model on the OpenWebText corpus at $\sim$ 400M parameters and find competitive perplexity vs. standard AR. They do not report a parameter-matched pure-AR baseline trained on the same data, nor a compute-matched control on the diffusion side.

Transfusion [12] trains a 7B single-transformer model with both an AR head (for text) and a continuous-diffusion head (for images). The work demonstrates joint training is feasible at scale and beats quantised-image-AR baselines. The training is not designed for parameter-matched comparison against pure-AR text models.

Janus and **Janus-Pro** [2, 10] train unified 7B models with separate visual-understanding and visual-generation encoders, with AR text generation. The architecture is closer to multi-modal AR than to AR+diffusion-text training.

8.2 Pre-trained AR $\rightarrow$ diffusion adaptation

DiffuLLaMA [4] adapts a pre-trained AR LLaMA backbone to diffusion mode via continued training. The reported result is ”competitive, not superior” vs. AR baselines. This is closely related to our compute-matched control but starts from a different initialisation; we train from scratch.

Fast-dLLM v2 [11] adapts a pre-trained AR model to block-diffusion mode and reports a 2.5$\times$ inference speedup at $<$ 1B tokens.

8.3 Continuous flow language models

ELF [1] demonstrates pure continuous flow language modelling. We replicated a small-scale flow variant (`ARFlowLM`) as part of our T1 ablations and found 0% accuracy at 60M-3k on GSM8K-dev — flow-only models do not compose with AR at our toy scale. We do not extend this branch.

8.4 Data-constrained discrete diffusion

DiFFPO [9] shows discrete diffusion outperforms AR in data-constrained settings via RL fine-tuning. Our data-efficiency curve replicates this regime effect in *from-scratch joint training* (no RL), at 200M scale, on math-reasoning rather than the language-modelling substrate DiFFPO used.

8.5 Mode-switching and interleaved inference

The original Sfumato planning document proposed a learned mode-router that selects among AR-extend, diffuse-current-block, re-diffuse-last-K-blocks at every sub-block boundary. We do not train such a router in this work; instead we test fixed mode-switching policies (e.g., AR-then-diffusion-revise) as the inference recipe. **SongBloom** [8] interleaves AR-sketch and diffusion-refine for music; the loop pattern is analogous but the domain is different.

9 Limitations and Future Work

9.1 Scale

All experiments are at 60M–300M parameters. The trade-off characterisation may shift at $\geq$ 1B; published works at 7B [4, 12] report different direction signals. We treat the 60M–300M characterisation as "small-scale evidence" and explicitly do not extrapolate.

9.2 Substrate

All training is on GSM8K-train ($\sim$ 4M tokens, math-reasoning domain). The data-efficiency crossover is substrate-specific; we make no claim that it generalises to broader corpora.

9.3 Eval metric

Downstream accuracy on GSM8K-dev is binomial-noise dominated at this scale (<6% correct out of 50 in most cells). We rely on continuous NLL metrics throughout, but downstream-task accuracy is the metric practitioners care about. Mode-switching at $n = 8$ tightens the CI but does not eliminate the noise.

9.4 Single hyperparameter trace

The α-schedule (linear $1.0 \to 0.5$), mask-ratio distribution (Uniform$(0.1, 0.9)$), and learning rate (6×10^{-4} with cosine decay) were picked from BD3-LMs defaults and not swept. A small α-schedule sweep is the natural follow-up.

9.5 Future work

1. Test the data-efficiency crossover on a non-math substrate (e.g., a $\sim$ 10M-token slice of a code corpus).
2. Extend the compute-matched control to AR-only-at-2$\times$ as well, to test "does composite also beat compute-matched pure-AR on its own axis at small data?"
3. Train a learned mode-router on a held-out per-window loss-vs-mode signal and test whether it beats fixed mode-switching policies at inference time.
4. Replicate at 1B from scratch to test whether the crossover shifts with scale.

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